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Roll No

EX-8402-GS
B.E. VIII Semester
Examination, December 2020
Grading System (GS)
Cellular Mobile Communication
(Elective - IV)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Explain the principle of operation of CDMA and derive the equation for the channel capacity in a single cell.
2. Why do we use 800 MHz for cellular mobile system.
3.
 - a) Explain point-to-point model with equations.
 - b) How will you increase the cell coverage for noise limited system?
4.
 - a) Describe diversity receivers.
 - b) Describe the process of cell splitting and sectorization.
5.
 - a) What is an umbrella-pattern effect? Write its benefits.
 - b) Write short note on:
 - i) Antenna Height
 - ii) Beam Tilting

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6. Discuss the traffic load on a set up channel and on voice channels. What do you understand by channel assignment?
7.
 - a) Draw the GSM architecture and explain each subsystem.
 - b) What is the call processing of mobile station in CDMA systems?
8. Write short notes on (any two)
 - a) TDD systems
 - b) PCN
 - c) UHF TV interference
 - d) Dropped call rate
 - e) Cordless phone
